

Bhabha scattering 2

For t -channel scattering amplitudes $\mathcal{M}_t$ we have

$$\mathcal{M}_t = \iint \bar{\psi}_2(y)(-ie\gamma^\mu)\psi_4(y)G(x,y)\bar{\psi}_3(x)(-ie\gamma_\mu)\psi_1(x) d^4x d^4y$$

where $G(x, y)$ is the photon propagator

$$G(x, y) = \frac{1}{(2\pi)^4} \int \frac{\exp(-ip \cdot (x - y))}{p^2 + i\epsilon} d^4p$$

The wave functions are

$$\begin{aligned}\psi_1(x) &= \sqrt{2EV}u_1 \exp(-ip_1 \cdot x) \\ \psi_2(x) &= \sqrt{2EV}v_2 \exp(ip_2 \cdot x) \\ \psi_3(x) &= \sqrt{2EV}u_3 \exp(-ip_3 \cdot x) \\ \psi_4(x) &= \sqrt{2EV}v_4 \exp(ip_4 \cdot x)\end{aligned}$$

where V is a normalization volume.

Substitute for the wave functions in $\mathcal{M}_t$ and factor the spinor products.

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \iint \exp(i(-p_2 + p_4) \cdot y) G(x, y) \exp(i(p_3 - p_1) \cdot x) d^4x d^4y$$

Substitute for $G(x, y)$.

$$\begin{aligned}\mathcal{M}_t &= -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \\ &\quad \frac{1}{(2\pi)^4} \iiint \exp(i(-p_2 + p_4) \cdot y) \frac{\exp(-ip \cdot (x - y))}{p^2 + i\epsilon} \exp(i(p_3 - p_1) \cdot x) d^4x d^4y d^4p\end{aligned}$$

Combine exponentials.

$$\begin{aligned}\mathcal{M}_t &= -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \\ &\quad \frac{1}{(2\pi)^4} \iiint \frac{\exp(i(-p_2 + p_4 + p) \cdot y) \exp(i(p_3 - p_1 - p) \cdot x)}{p^2 + i\epsilon} d^4x d^4y d^4p\end{aligned}$$

Solve the integral over x .

$$\int \exp(i(p_3 - p_1 - p) \cdot x) d^4x = (2\pi)^4 \delta^4(p_3 - p_1 - p)$$

Solve the integral over y .

$$\int \exp(i(-p_2 + p_4 + p) \cdot y) d^4y = (2\pi)^4 \delta^4(-p_2 + p_4 + p)$$

Hence

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1)(2\pi)^4 \int \frac{\delta^4(-p_2 + p_4 + p)\delta^4(p_3 - p_1 - p)}{p^2 + i\epsilon} d^4p$$

The integral over p vanishes except for

$$p = p_2 - p_4 = p_3 - p_1$$

Hence

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1)(2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_3 - p_1)^2}$$

For momentum vectors

$$p_1 = \begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}_{e^- \rightarrow} \quad p_2 = \begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}_{\leftarrow e^+} \quad p_3 = \begin{pmatrix} E \\ p \sin \theta \cos \phi \\ p \sin \theta \sin \phi \\ p \cos \theta \end{pmatrix}_{e^- \nearrow} \quad p_4 = \begin{pmatrix} E \\ -p \sin \theta \cos \phi \\ -p \sin \theta \sin \phi \\ -p \cos \theta \end{pmatrix}_{\nwarrow e^+}$$

we have $\delta^4(p_4 + p_3 - p_2 - p_1) = \delta^4(0) = 1$ hence

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \frac{(2\pi)^4}{t}$$

where t is the Mandelstam variable

$$t = (p_1 - p_3)^2$$

The s -channel scattering amplitudes $\mathcal{M}_s$ have ψ_2 and ψ_3 interchanged relative to $\mathcal{M}_t$.

$$\mathcal{M}_t = \iint \bar{\psi}_3(y)(-ie\gamma^\mu)\psi_4(y)G(x, y)\bar{\psi}_2(x)(-ie\gamma_\mu)\psi_1(x) d^4x d^4y$$

It follows that

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2}(\bar{u}_3\gamma^\mu v_4)(\bar{v}_2\gamma_\mu u_1)(2\pi)^4 \int \frac{\delta^4(p_3 + p_4 + p)\delta^4(-p_2 - p_1 - p)}{p^2 + i\epsilon} d^4p$$

The integral over p vanishes except for

$$p = -p_3 - p_4 = p_1 + p_2$$

Hence

$$\mathcal{M}_s = -\frac{e^2}{(2EV)^2}(\bar{u}_3\gamma^\mu v_4)(\bar{v}_2\gamma_\mu u_1) \frac{(2\pi)^4}{s}$$

where s is the Mandelstam variable

$$s = (p_1 + p_2)^2$$

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